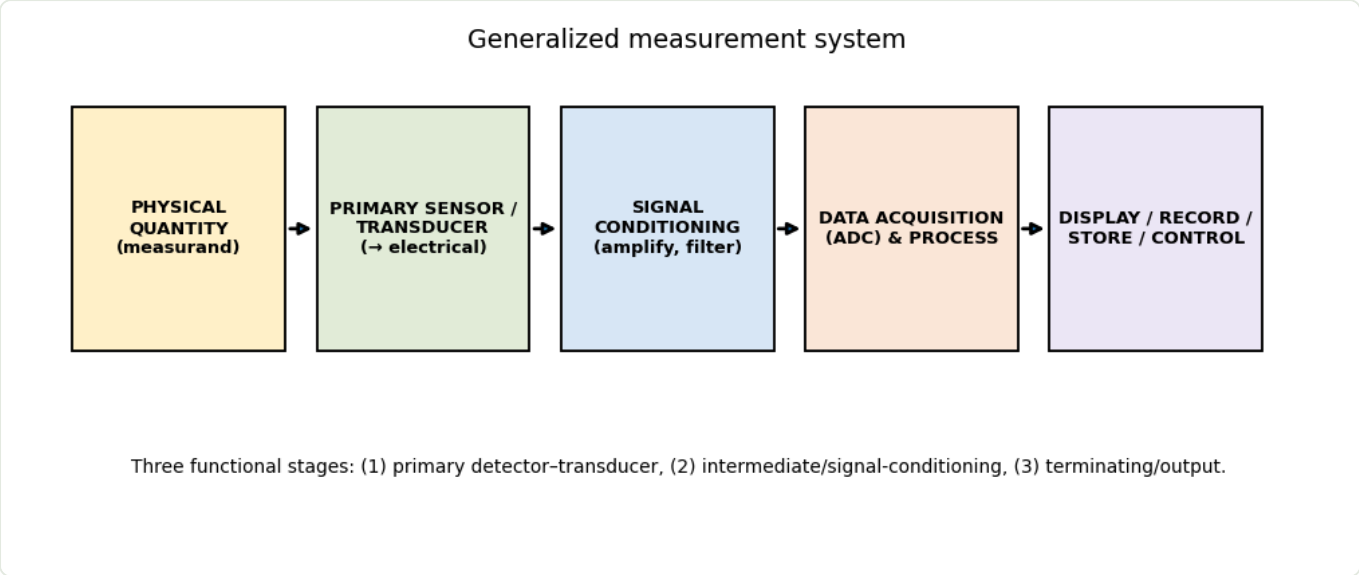
UNIT 9 · TOPIC 1

Measurement Systems, Sensors and Transducers

Syllabus topic: Mechanical measurements; sensors and transducers: types and applications

The generalized measurement system



Generalized measurement system

1. **PRIMARY DETECTOR–TRANSDUCER** stage — senses the measurand and converts it, usually to an electrical signal.
2. **INTERMEDIATE / SIGNAL-CONDITIONING** stage — amplifies, filters, modifies the signal (Topic 3).
3. **TERMINATING / OUTPUT** stage — indicates, records, stores or controls.

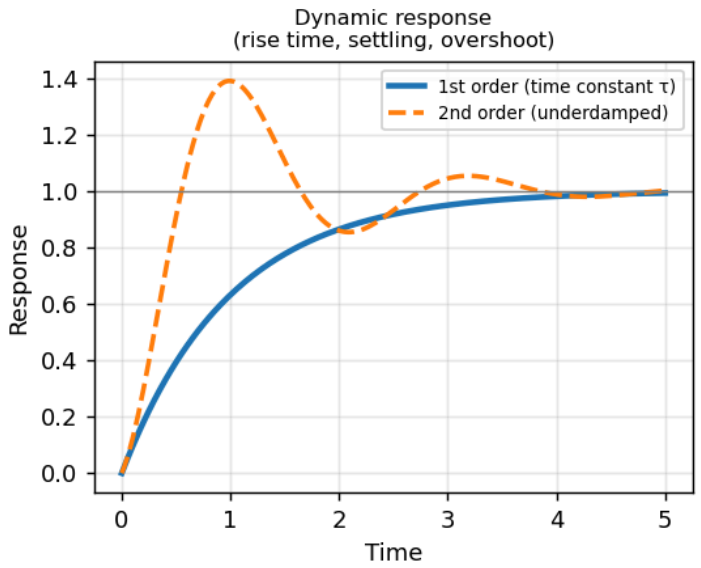
In farm-machinery research these chains measure draft, torque, force, pressure, speed, vibration, temperature, flow, etc. (Units 2–6).

Sensor vs transducer; active vs passive

- A **SENSOR** detects the physical quantity; a **TRANSDUCER** converts one form of energy into another (usually the measurand into an electrical output). Every transducer senses, but a transducer specifically delivers a convertible (electrical) signal.

Type	How it works	Examples
ACTIVE (self-generating)	Generates its own output	Thermocouple, piezoelectric
PASSIVE	Needs external excitation; changes a parameter (resistance/capacitance/inductance)	Strain gauge, LVDT, RTD

Characteristics of instruments



CHARACTERISTICS & ERRORS

STATIC: accuracy, precision, sensitivity, linearity, range/span, resolution, hysteresis, drift, threshold, dead zone.

DYNAMIC: response time, time constant τ, natural frequency, damping, bandwidth, rise/settling time, overshoot.

ERRORS: SYSTEMATIC (bias — calibration, loading, environment; correctable) vs RANDOM (scatter; reduce by averaging). Accuracy = closeness to true value; Precision = repeatability.

Dynamic response of first- and second-order instruments, and the characteristics/errors checklist

Static characteristic (steady input)	Meaning
ACCURACY	Closeness to the true value
PRECISION	Repeatability / scatter of readings
SENSITIVITY	Output change ÷ input change
LINEARITY; RANGE/SPAN	—
RESOLUTION	Smallest detectable change in input
THRESHOLD; dead zone	Minimum input before any response
HYSTERESIS	Difference between loading (increasing) and unloading (decreasing) readings
DRIFT	Slow change of output over time

Dynamic characteristics (time-varying input): response time, TIME CONSTANT τ (first-order systems), NATURAL FREQUENCY and DAMPING RATIO (second-order — e.g., a diaphragm gauge), bandwidth, rise/settling time, overshoot.

Errors and calibration

Error type	Nature	Remedy
SYSTEMATIC (bias)	Consistent — calibration, LOADING effect (the instrument disturbs what it measures), zero/scale and environmental errors	Correct by calibration
RANDOM	Scatter unpredictably	Repeated readings and averaging

- Overall accuracy is often stated as ± % of full scale.

- Calibration against a known standard traceable to national standards (NPL India) underpins all measurement.

Practice MCQs

Q1. The three functional stages of a measurement system are transducer, signal conditioning and:

(a) Battery
(b) Terminating/output stage
(c) Motor
(d) Pump

Q2. A device that converts a physical quantity into an electrical signal is a:

(a) Actuator
(b) Transducer
(c) Relay
(d) Rectifier

Q3. A thermocouple is an example of a/an:

(a) Passive transducer
(b) Active (self-generating) transducer
(c) Actuator
(d) Amplifier

Q4. A strain gauge is a/an:

(a) Active transducer
(b) Passive transducer (needs excitation)
(c) Self-generating
(d) Digital display

Q5. Accuracy of an instrument refers to:

(a) Repeatability
(b) Closeness to the true value
(c) Smallest division
(d) Speed

Q6. Precision refers to:

(a) Closeness to true value
(b) Repeatability of readings
(c) Range
(d) Sensitivity

Q7. The smallest change in input that an instrument can detect is its:

(a) Range
(b) Resolution
(c) Hysteresis
(d) Drift

Q8. Sensitivity is defined as:

(a) Input ÷ output
(b) Output change ÷ input change
(c) Range × span
(d) Error ÷ reading

Q9. The difference in readings for increasing vs decreasing input is:

(a) Drift
(b) Hysteresis
(c) Threshold
(d) Bias

Q10. Systematic (bias) errors are:

(a) Random and unpredictable
(b) Consistent and correctable by calibration
(c) Reduced only by averaging
(d) Never present

Q11. Random errors are best reduced by:

(a) Calibration
(b) Repeated readings and averaging
(c) Ignoring
(d) A bigger scale

Q12. The error where the instrument disturbs the measured quantity is the:

(a) Parallax error
(b) Loading effect
(c) Zero error
(d) Quantisation error

Q13. A first-order instrument's speed of response is characterised by its:

(a) Gauge factor
(b) Time constant τ
(c) Range
(d) Span

Q35. Calibration of an instrument is done against a:

(a) Guess
(b) Known standard traceable to national standards
(c) Similar faulty unit
(d) Colour chart

Answers

Q1 — (b) Terminating/output stage
Q2 — (b) Transducer
Q3 — (b) Active (self-generating) transducer — *Active = self-generating (thermocouple, piezo).*
Q4 — (b) Passive transducer (needs excitation) — *Passive = needs excitation (strain gauge, LVDT, RTD).*
Q5 — (b) Closeness to the true value
Q6 — (b) Repeatability of readings
Q7 — (b) Resolution
Q8 — (b) Output change ÷ input change
Q9 — (b) Hysteresis
Q10 — (b) Consistent and correctable by calibration
Q11 — (b) Repeated readings and averaging
Q12 — (b) Loading effect
Q13 — (b) Time constant τ
Q35 — (b) Known standard traceable to national standards

Sources: Doebelin — Measurement Systems; Beckwith — Mechanical Measurements; Sawhney — Electrical & Electronic Measurements; HBM/Omega strain-gauge technical notes (as given in the source notes, July 2026); ASAE instrumentation standards; NPTEL measurement lectures.